

THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 4

1. Find the partial-fraction decomposition of the following rational functions:

(a) i.

$$\frac{1}{(x+5)(x+4)}$$

ii.

$$\frac{x+7}{x^2+3x+2}$$

iii.

$$\frac{2x^2-9x-35}{x^3+2x^2-5x-6}$$

(b) i.

$$\frac{x^2+2}{(x+2)^2(x+3)}$$

ii.

$$\frac{4x}{x^3+4x^2+5x+2}$$

(c) i.

$$\frac{x^2-1}{x(x^2+1)}$$

ii.

$$\frac{x-3}{x^3+3x}$$

iii.

$$\frac{2x}{(x+1)^2(x^2+1)}$$

(d) i.

$$\frac{x^2+1}{x^2-1}$$

ii.

$$\frac{4x^3+10x+4}{x(2x+1)}$$

iii.

$$\frac{2x^4+3x^2+1}{x^2+3x+2}$$

2. (a) Solve each of the following quadratic inequalities, in each case, illustrating the solution set on the number line.

i.

$$2x^2+3x \geq 3$$

iii.

$$x^2+2x \leq 8$$

ii.

$$x^2-2x-3 < 0$$

iv.

$$x^2-3x-4 \neq 0$$

(b) Sketch each of the following quadratic functions. In each case, state the set of value(s) of x for which $f(x) > 0$, $f(x) \leq 0$.

i.

$$f(x) = 2x^2 + 3x - 3$$

iii.

$$f(x) = 2x^2 - 4x + 5$$

ii.

$$f(x) = -x^2 - 4x + 1$$

iv.

$$f(x) = -2x^2 - 3x + 2$$

(c) Solve each of the following cubic inequalities, in each case, illustrating the solution set on the number line.

i.

$$4x^3 - x < 6 - 9x^2$$

iii.

$$x^3 - x^2 - 9x + 9 \leq 0$$

ii.

$$(x + 4)(x - 2)(x - 7) > 0$$

iv.

$$4x^3 - 8x^2 - x + 2 \neq 0$$

(d) Sketch each of the following cubic functions. In each case, state the set of value(s) of x for which $f(x) > 0$, $f(x) \leq 0$.

i.

$$f(x) = 4x^3 + 9x^2 - x - 6$$

iii.

$$f(x) = x^3 - 5x^2 + 3x + 9$$

ii.

$$f(x) = x^3 - 5x^2 - 22x + 56$$

iv.

$$f(x) = x^4 + 4x^3 - 12x^2$$

3. (a) Solve each of the following rational inequalities, in each case, illustrating the solution set on the number line.

i.

$$\frac{x - 1}{x + 2} \leq 0$$

v.

$$\frac{x}{x + 1} \leq \frac{x}{x + 2}$$

ii.

$$\frac{x + 2}{x - 1} \geq 1$$

vi.

$$x(x - 1) > \frac{x - 1}{x}$$

iii.

$$\frac{x^2 - 9}{x^2 - 1} < 0$$

vii.

$$\frac{x - 3}{x + 4} \geq \frac{x + 2}{x - 5}$$

iv.

$$\frac{x^2 + 5x + 6}{x^2 - 4x - 5} \leq 0$$

(b) For each of the following rational functions, find/state the domain and the range. Hence sketch the graph.

i.

$$f(x) = \frac{2 - x}{x + 4}$$

iii.

$$f(x) = \frac{x + 1}{x - 1}$$

ii.

$$f(x) = \frac{3x + 6}{x - 1}$$

iv.

$$f(x) = \frac{2x + 5}{x - 1}$$